

Time & Space Complexity (Assignment Solutions)

Solution :

- a. Option A -> Time complexity = $O(n \cdot \log n)$
 In the loop, j keeps doubling till it is less than or equal to n. Several times, we can double a number till it is less than n would be $\log(n)$.
 Let's take the examples here.
 for $n = 16, j = 2, 4, 8, 16$
 for $n = 32, j = 2, 4, 8, 16, 32$
 So, j would run for $O(\log n)$ steps.
 i runs for $n/2$ steps.
 So, total steps = $O(n/2 * \log(n)) = O(n \cdot \log n)$
- b. Option C -> Time complexity = $O(\log kn)$
 Because loops for the $kn-1$ times, so after taking log it becomes $\log kn$.
- c. Option B. -> false
 The Big-O notation provides an asymptotic comparison in the running time of algorithms. For $n < n_0$, algorithm A might run faster than algorithm B, for instance.
- d. Time complexity - $O(\sqrt{n})$ //square root of n
 Space complexity - $O(1)$
- e. Time complexity - $O(n^2)$
 Space complexity - $O(1)$